Code No. 3413

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE, AI&ML) IV-Semester (MAIN) Examinations July-2025

SIGNALS AND SYSTEMS

: 3 hours

Max. Marks : 60

e : i) *Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) *Answers must be written in same order as they occur in the Question Paper.*
iii) ***Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	Determine whether the system : $y(t) = x^2(t) + 3x(t-1)$ is i) linear or nonlinear ii) Causal or non-causal	2	1	1
b)	Express trigonometric Furrier series coefficients in terms of exponential Fourier series.	2	2	2
c)	State and prove initial value property of Laplace Transform.	2	3	3
d)	What happens when a signal is under-sampled?	2	5	4
e)	Define Region of Convergence (RoC) and explain its importance in Z-transform.	2	4	5
f)	Verify the time scaling property of Fourier Transform for $x(t) = e^{-t}u(t)$	2	2	3
2. a)	Given $x(t) = e^{-t}u(t)$ perform and sketch the following: i) $x(t-2)$ ii) $x(2t)$ iii) $x(-t)$ iv) $x(t+2)$	6	2	1
b)	Discuss the following i) Static and dynamic systems ii) Lumped and distributed parameter systems	6	3	1
3. a)	Find the trigonometric Fourier series coefficients for a given half-wave rectified sine wave.	6	4	2
b)	Compute the response of an LTI system with impulse response $h(t) = e^{-t}u(t)$ and input $x(t) = cos(t)$ using Fourier series.	6	3	2

(P.T.O.)

4. a) Determine Fourier Transform of the following signals 8 4 4

i) Rectangular function

ii) Signum function

b) Solve the differential equation $\frac{dy}{dt}+5y(t)=x(t)$ using Laplace transforms. 4 5 3

5. a) A continuous-time signal $x(t)=cos(3\pi t)$ is sampled $fs=2Hz$ 12 4 4

i) Find the sampled signal x[n]

ii) Determine if aliasing occurs

iii) Plot both the original and aliased frequency

6. a) Write and prove properties of DTFT 5 5

i) Parseval's Theorem ii) Time Differentiation

iii) Convolution iv) Time scaling

b) Find the inverse Z-transform 5 3 5

$$X(z)=\frac{1+3z^{-1}}{1+3z^{-1}+2z^{-2}},\ |z|>2$$

7. a) Find the inverse Laplace transform of 5 2 3

$$X(s)=\frac{5(s+5)}{s(s+3)(s+7)}:Re(s)>-3$$

b) Prove that the sequence $x(n)=a^n u(n)$) and $x(n)=-a^n u(-n-1)$ have the same $X(z)$ and differ only in RoC. Also plot their RoC. 7 4 5

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